

DOCUMENT INDEX — Master Control Register

OCP Maintenance AI MVP

Purpose: This is the single entry point for ALL project knowledge. Agents, skills, and developers MUST consult this index FIRST before reading full documents. Each entry provides a summary and routing guidance to minimize token consumption.

Rule: Read the SUMMARY document first. Only read the FULL document if deeper detail is needed.

HOW TO USE THIS INDEX

1. **Identify your topic** from the categories below
 2. **Read the SUMMARY document** (architecture/ref-XX) — typically 200-400 lines, covers 90% of needs
 3. **Only if needed**, read the FULL SOURCE document for raw detail
 4. **For schemas/rules**, always refer to `gemini.md` as the single source of truth
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TIER 1: PROJECT GOVERNANCE (Read First, Always)

ID	Document	Path	Purpose	When to Read
GOV-01	gemini.md	<code>gemini.md</code>	PROJECT CONSTITUTION. Data schemas, behavioral rules, architectural invariants, integration registry. This is LAW.	ALWAYS read first for any schema, rule, or constraint question
GOV-02	DOCUMENT_INDEX.md	<code>DOCUMENT_INDEX.md</code>	THIS FILE. Master navigation. Routes to correct document.	When starting any new task or research
GOV-03	task_plan.md	<code>task_plan.md</code>	Phases, goals, checklists. Current project status.	When checking what to do next
GOV-04	progress.md	<code>progress.md</code>	Session log. What was done, errors, tests.	When resuming work or checking history
GOV-05	findings.md	<code>findings.md</code>	Research discoveries, client context summary, technology decisions.	When needing project context quickly

TIER 2: REFERENCE DOCUMENTS (Summaries — Read These)

Domain Knowledge

ID	Document	Path	Covers	Lines	Token Est.
REF-01	Maintenance Strategy Methodology	architecture/ref-01-maintenance-strategy-methodology.md	Full RCM process: 5 phases, criticality matrix, decision tree, task definition, work packaging, frequency selection	~400	~3K
REF-02	R8 Data Model & Entities	architecture/ref-02-r8-data-model-entities.md	All R8 entity schemas (12+ entities), field definitions, code tables, ERD, library system	~350	~2.5K
REF-04	Quality Validation Rules	architecture/ref-04-quality-validation-rules.md	6-stage QA process, 40+ field-level validation rules, MSO checklists, lessons learned	~350	~2.5K
REF-07	Work Instruction Templates	architecture/ref-07-work-instruction-templates.md	WI structure, 4 WP type templates, naming conventions, material kits, SAP mapping	~250	~2K

Technical Integration

ID	Document	Path	Covers	Lines	Token Est.
REF-03	SAP PM Integration	architecture /ref-03-sap- pm- integration.m d	SAP upload templates (3 files, 64 fields total), cross-reference model, field mappings, upload process, SAP transactions	~300	~2.5K
REF-06	Software Architecture Vision	architecture /ref-06- software- architecture- vision.md	Second Brain vision, full tech stack (20+ technologies), 10 AI use cases, UX principles, competitive landscape	~300	~2.5K

Client & Business

ID	Document	Path	Covers	Lines	Token Est.
REF-05	Client Context OCP	architecture /ref-05- client- context- ocp.md	OCP pain points, MVP scope (3+1 modules), timeline (16-24 weeks), success metrics, risks, data requirements	~250	~2K

User Guide

ID	Document	Path	Covers	Lines	Token Est.
REF-08	Step-by-Step User Guide	architecture/ref-08-user-guide-step-by-step.md	Complete step-by-step for all 4 modules, daily workflows, workshop workflow, emergency workflow	~400	~3K

Standards & Compliance

ID	Document	Path	Covers	Lines	Token Est.
REF-09	ISO 55002 Compliance Mapping	architecture/ref-09-iso-55002-compliance-mapping.md	ISO 55002:2018 clause-by-clause mapping, compliance scorecard (80%, up from 73%), gap analysis with 5 closed gaps, 9 remaining gaps	~290	~2.5K

Competitive Intelligence & Strategy

ID	Document	Path	Covers	Lines	Token Est.
REF-10	GECAMIN	architecture	22+ GECAMIN	~350	~3K
	MAPLA 2024 Cross-Reference	/ref-10- gecamin- cross- reference.md	presentations analyzed, myRIAM competitor analysis, competitive positioning matrix, validated claims, 9 recommendatio ns		
REF-11	Neuro- Arquitectura Integrated Review	architecture /ref-11- neuro- arquitectura- review.md	6 behavioral science pillars mapped to 4 modules, implementation checklists, bias mitigation, ISO 55002 integration, GECAMIN cross-validation	~350	~3K
REF-12	Final Strategic Recommendati ons	architecture /ref-12- strategic- recommendatio ns.md	10 strategic recommendatio ns, 3-layer competitive moat, implementation priority matrix, risk assessment, Phase 0 plan	~350	~3K

GFSN Maintenance Methodology (Gold Fields Salares Norte)

ID	Document	Path	Covers	Lines	Token Est.
REF-13	Maintenance Management Manual	architecture /ref-13- maintenance- manual- methodology.m d	GFSN maintenance governance, FMECA, criticality (3-level Alto/Moderado /Bajo), technical hierarchy naming, RBI, RCM process, improvement techniques (OCR, Pareto, Jack-Knife, Weibull, LCC, MoC), 8 KPIs, organizational structure	~350	~3K
REF-14	Planning & Scheduling Procedure	architecture /ref-14- planning- scheduling- procedure.md	6-stage weekly cycle, two work streams (preventive from Plan Matriz + corrective from Avisos), priority matrix, SAP WO status flow, work package contents, 11 KPIs with targets, 7 roles, weekly scheduling meeting structure	~240	~2K
REF-15	Defect Elimination Procedure	architecture /ref-15- defect- elimination- procedure.md	5-stage DE process (Identify→Prioritize→Analyze→Implement→Control), 5W+2H	~300	~2.5K

ID	Document	Path	Covers	Lines	Token Est.
			method, RCA Cause-Effect (Ishikawa), 5P's evidence framework, 3- level root cause (Physical→Hum an→Latent), solution prioritization matrix, 5 KPIs		
REF-16	Asset Criticality Analysis Procedure	architecture /ref-16- criticality- analysis- procedure.md	Semi- quantitative method, 6 consequence factors (3 economic + 3 non-economic), 5×5 matrix, 3 criticality levels (Alto 19-25, Moderado 8-18, Bajo 1-7), ALARP zones, evaluation assumptions, comparison with current 4- class system	~200	~2K
REF-17	Methodology Gap Analysis & Roadmap	architecture /ref-17- methodology- gap-analysis- roadmap.md	20 gaps identified, detailed analysis per domain, proposed architectures for scheduling engine + RCA engine, 11 planning KPIs, 5 DE KPIs, technical roadmap (Phases 4A/4B/5), 7	~400	~3.5K

ID	Document	Path	Covers	Lines	Token Est.
			decision points, 11 new MCP tools, agent prompt updates		

Architecture & Blueprint

ID	Document	Path	Covers	Lines	Token Est.
BP-01	Blueprint	architecture /BLUEPRINT.m d	Architecture diagram, 5-step build sequence, safety flow, trilingual approach, file structure, dependencies	~350	~2.5K

TIER 3: SOURCE DOCUMENTS (Full Detail — Read Only When Needed)

Client Context (Original)

ID	Document	Path	Format	Pages	When to Read
SRC-01	RFI Presentation	CLIENT CONTEXT/RFI AI-Powered Asset Management and Maintenance Solution.pdf	PDF	17 slides	Need exact RFI wording, investment framework, or specific slide content
SRC-02	Data Requirements	CLIENT CONTEXT/OCP_AI_Maintenance _Solution_Proposal.pdf	PDF	8 pages	Need exact field specifications for 15 data categories or folder structure
SRC-03	Meeting Transcripts	CLIENT CONTEXT/MEETINGS/	DOCX/PDF	Multiple	Need exact meeting context, decisions, or action items

R8 Software Documentation (Original)

ID	Document	Path	Format	When to Read
SRC-09	Failure Modes Lookup Table (AUTHORITATIVE)	MAINTENANCE STRATEGY.../Failure Modes (Mechanism + Cause).xlsx	XLSX (72 rows)	MANDATORY for any FMEA or failure mode work. Contains the ONLY 72 valid Mechanism+Cause combinations. ALL failure modes MUST reference this table. Any agent or assistant working with failure modes MUST validate against this table.
SRC-10	R8 Tactics Library Config	MAINTENANCE STRATEGY.../R8-software-tactics-library-configuration-procedure.pdf	PDF (147p)	Need exact R8 field names, screenshots, or configuration steps
SRC-11	R8 QA Guideline	MAINTENANCE STRATEGY.../R8-software-maintenance-strategy-quality-analysis-guideline.pdf	PDF	Need exact QA rule wording or examples
SRC-12	Maintenance Tactics Guideline	MAINTENANCE STRATEGY.../main-tenance-strategy-tactics-developmenet-Guideline v0.4.pdf	PDF	Need exact Anglo American methodology text, criticality matrix details
SRC-13	Maintenance Tactics Process Map	MAINTENANCE STRATEGY.../main-tenance-strategy-	PDF	Need exact process flow diagrams

ID	Document	Path	Format	When to Read
		tactics-development-Process-Map.pdf		
SRC-14	R8 Asset Tactics Process	MAINTENANCE STRATEGY.../R8-asset-maintenance-tactics-development-process.docx	DOCX	Need exact workshop procedure, 3 approaches detail
SRC-15	RCM Documentation	MAINTENANCE STRATEGY.../rcm-reliability-centred-maintenance.docx	DOCX	Need exact RCM theory, simplified criticality method
SRC-16	R8 Deployment	MAINTENANCE STRATEGY.../R8 Deployment - Implementation.pdf	PDF	Need exact 6-phase deployment procedure
SRC-17	R8 Maintenance Plans Optimization	MAINTENANCE STRATEGY.../R8-Maintenance-plans-Optimization.pdf	PDF	Need exact optimization algorithms or methods
SRC-18	R8 Flowcharts	MAINTENANCE STRATEGY.../R8-software-deployment-flow-charts.pdf	PDF	Need exact R8 process flowcharts
SRC-19	R8 Tactics Library Procedure	MAINTENANCE STRATEGY.../R8-software-tactics-library-configuration-procedure.pdf	PDF	Need exact library configuration steps

ID	Document	Path	Format	When to Read
SRC-20	R8 Integration Master Plan	MAINTENANCE STRATEGY.../Ryls on8-software-Integration-Master-Plan.docx	DOCX	Need exact R8 module descriptions or integration details
SRC-21	MSO Process	MAINTENANCE STRATEGY.../R8-maintenance-strategy-optimization-process.pptx	PPTX	Need exact MSO step-by-step with screenshots
SRC-22	MSO Information	MAINTENANCE STRATEGY.../MSO informacion.docx	DOCX	Need exact workshop preparation questions (in Spanish)
SRC-23	MSO Checklist	MAINTENANCE STRATEGY.../Check_list_MSO.xlsx	XLSX	Need exact checklist items with columns
SRC-24	QA Flowchart	MAINTENANCE STRATEGY.../main tenance-strategy-development-quality-management-flowchart.xlsx	XLSX	Need exact QA flowchart steps, lessons learned

SAP Integration (Original)

ID	Document	Path	Format	When to Read
SRC-30	SAP EAM Presentation	MAINTENANCE STRATEGY.../SAP EAM (PM).pptx	PPTX	Need exact SAP PM structure screenshots or entity details
SRC-31	Setting up Plans in SAP	MAINTENANCE STRATEGY.../Setting-up-Mtce-Plans-in-SAP-using R8.pdf	PDF	Need exact R8→SAP mapping procedure with screenshots
SRC-32	SAP Maintenance Item Template	MAINTENANCE STRATEGY.../sap-upload-sheets-template-.../Maintenance Item.xlsx	XLSX	Need exact SAP upload field structure
SRC-33	SAP Task List Template	MAINTENANCE STRATEGY.../sap-upload-sheets-template-.../Task List.xlsx	XLSX	Need exact SAP task list operation fields
SRC-34	SAP Work Plan Template	MAINTENANCE STRATEGY.../sap-upload-sheets-template-.../Work Plan.xlsx	XLSX	Need exact SAP maintenance plan fields

Work Instruction Examples (Original)

ID	Document	Path	Format	When to Read
SRC-40	Anglo American WP Templates	MAINTENANCE STRATEGY.../main tenance-work- instruction- template- examples/Anglo American Coal Work package templates/	DOCX	Need exact WP template formatting, fields, examples
SRC-41	WI Template Examples	MAINTENANCE STRATEGY.../main tenance-work- instruction- template- examples/	DOCX/DOC/PDF	Need exact work instruction formatting from various sources

Software Development Context (Original)

ID	Document	Path	Format	When to Read
SRC-50	Antigravity Second Brain	SOFTWARE DEVELOPMENT CONTEXT/Antigravity second brain.docx.md	MD	Need exact tech stack benchmarks, 90-day plan, risk mitigations
SRC-51	Second Brain Architecture	SOFTWARE DEVELOPMENT CONTEXT/Arquitectura del Second Brain Corporativo.md	MD	Need exact 4-pillar architecture, Nexus prototype JSON
SRC-52	Technical Architecture Deep Dive	SOFTWARE DEVELOPMENT CONTEXT/Diseño Arquitectura Segundo Cerebro Corporativo1.docx.md	MD	Need exact SQL DDL, Python code, Supabase schema, PII pipeline
SRC-53	Second Brain Model	SOFTWARE DEVELOPMENT CONTEXT/El Modelo del _Second Brain_ Corporativo...md	MD	Need exact prototype case study, implementation roadmap
SRC-54	Mining AI & Digitalization	SOFTWARE DEVELOPMENT CONTEXT/Gestión minera_ IA, NO-Code y digitalización..md	MD	Need exact ROI figures, Airtable mapping, KPI framework
SRC-55	Neuro-Architecture UX	SOFTWARE DEVELOPMENT CONTEXT/Neuro-Arquitectura Organizacional..md	MD	Need exact cognitive bias table, nudge taxonomy, CLT guidelines

ID	Document	Path	Format	When to Read
SRC-56	10 AI Prototypes	SOFTWARE DEVELOPMENT CONTEXT/Prototip ado IA para Gestión de Activos Industriales.md	MD	Need exact use case details, JSON samples, department-specific requirements
SRC-57	Megaproject Software	SOFTWARE DEVELOPMENT CONTEXT/Software Megaproyectos_ Ingeniería a Operación.md	MD	Need exact software comparison tables, CFIHOS standard, implementation horizons
SRC-58	AI Startup vs Incumbents	SOFTWARE DEVELOPMENT CONTEXT/Startup AI vs. Incumbents_ Megaproyectos.m d	MD	Need exact startup profiles, GraphRAG details, Rust architecture rationale

GECAMIN MAPLA 2024 Conference (Original)

ID	Document	Path	Format	When to Read
SRC-70	myRIAM SYSTEM (Guayacán Solutions)	GECAMIN MAPLA 2024 PPTS/S11 15_10 Sergio Garcia_Sistema de gestión de activos y mantenimiento basado en IA Generativa, myRIAM SYSTEM.pdf	PDF (14p)	Need exact myRIAM architecture, competitor feature comparison
SRC-71	AI Detector — Crusher Anomalies	GECAMIN MAPLA 2024 PPTS/S11 14_50 Rodrigo Vergara_AI Detector...pdf	PDF (12p)	Need exact AI anomaly detection methodology for crushers
SRC-72	Chatbot Experto (SQM)	GECAMIN MAPLA 2024 PPTS/S14 16_50 Jorge Martinez_Chatbot Experto...pdf	PDF (11p)	Need exact chatbot question taxonomy, Copilot Studio architecture
SRC-73	Gabinete Smart / Digital Twin (I&T)	GECAMIN MAPLA 2024 PPTS/S14 17_10 Hugo Barrientos_Gabin ete smart...pdf	PDF (14p)	Need exact digital twin ecosystem, IIoT sensor architecture
SRC-74	GenAI in Hydraulic Shovels (Centinela)	GECAMIN MAPLA 2024 PPTS/S5 14_10 Erick Parra_VIDEO_AI Generativa en palas hidráulicas PC5500.pdf	PDF (18p)	Need exact MTTR reduction methodology via GenAI
SRC-75	AI ConMon Optimization	GECAMIN MAPLA 2024 PPTS/S5 14_50 Patricio Ortiz_Optimizaci ón del análisis	PDF (17p)	Need exact efficiency gains (55% time reduction), AI

ID	Document	Path	Format	When to Read
		de monitoreo...pdf		classification accuracy
SRC-76	Predictive Maintenance AI (U. Chile)	GECAMIN MAPLA 2024 PPTS/S3 11_50 Vivana Meruane_Transfor mando el Mantenimiento Predictivo...pd f	PDF (27p)	Need exact academic framework for anomaly detection, diagnosis, RUL
SRC-77	797F Engine Failure Prediction	GECAMIN MAPLA 2024 PPTS/S7 17_10 Jean Campos_Modelo de predicción de falla...pdf	PDF (15p)	Need exact SVM model, 83% accuracy metrics, prediction window
SRC-78	Asset Health Index (SQM Lithium)	GECAMIN MAPLA 2024 PPTS/S6 14_50 Cristian Ramirez_Índice de salud de activos...pdf	PDF (13p)	Need exact health index variables, criticality-backlog- strategy formula
SRC-79	Codelco Unified Data	GECAMIN MAPLA 2024 PPTS/S9 09_00 Jorge Hidalgo_Desempeñ o de activos a través del dato único.pdf	PDF (15p)	Need exact Codelco scale (15,638 equipment), centralized management model
SRC-80	SOMA Maintenance Operating System (Codelco)	GECAMIN MAPLA 2024 PPTS/S13 17_10 Carlo Lobiano_Mejora en la implementación del sistema operativo...pdf	PDF (20p)	Need exact change management challenges, operations- maintenance collaboration

ID	Document	Path	Format	When to Read
SRC-81	Mining 4.0 Shutdown Optimization (Andes Agile)	GECAMIN MAPLA 2024 PPTS/S16 09_40 Roberto Verdugo_Minería 4.0...pdf	PDF (17p)	Need exact cloud-based shutdown optimization methodology
SRC-82	All GECAMIN Presentations (57 PDFs)	GECAMIN MAPLA 2024 PPTS/	PDF (57 files)	Browse full conference proceedings for additional competitive intelligence

GFSN Methodology Documents (Original)

ID	Document	Path	Format	When to Read
SRC-90	GFSN Maintenance Management Manual	MAINTENANCE STRATEGY.../main tenance-manual--GFSN01-DD-EM-0000-MN-00001-...pdf	PDF (104p)	Need exact GFSN process map, FMECA methodology, improvement techniques, organizational structure
SRC-91	GFSN Planning & Scheduling Procedure	MAINTENANCE STRATEGY.../plan ning-scheduling-procedure--GFSN01-DD-EM-0000-PT-00006-...pdf	PDF (34p)	Need exact weekly scheduling cycle, SAP WO status flow, work package requirements, KPI formulas
SRC-92	GFSN Defect Elimination Procedure	MAINTENANCE STRATEGY.../defect-elimination-procedure--GFSN01-DD-EM-0000-PT-00005-...pdf	PDF (39p)	Need exact RCA methodology, 5P's framework, Cause-Effect diagram steps, solution prioritization
SRC-93	GFSN Asset Criticality Analysis Procedure	MAINTENANCE STRATEGY.../asset-criticality-analysis-procedure--GFSN01-DD-EM-0000-PT-00001-...pdf	PDF (16p)	Need exact 6-factor consequence evaluation, 5×5 matrix, criticality level definitions

Video Tutorials (Original)

ID	Document	Path	Format	When to Read
SRC-60	R8 Video Tutorials	MAINTENANCE STRATEGY.../Video-tutorial-usage-R8/	MP4 (11 files)	Cannot be read by AI. Reference only for human training.

TOPIC ROUTING TABLE

Use this table to find the right document for any question:

Topic	Quick Answer	Summary Doc	Full Detail
Data schemas (any entity)	gemini.md §3 or §9	-	-
Behavioral rules	gemini.md §4	-	-
RCM decision tree	REF-01 §3.4	REF-01	SRC-12, SRC-15
Criticality assessment	REF-01 §2	REF-01	SRC-12
Failure mode analysis (FMEA)	REF-01 §3.3	REF-01	SRC-09 (MANDATORY), SRC-10, SRC-14
Valid FM combinations (Mechanism+Cause)	SRC-09 (72 combos)	REF-01 §3.3	SRC-09 IS THE ONLY SOURCE OF TRUTH
Task naming conventions	REF-01 §4.2	REF-01	SRC-11
R8 entity fields	REF-02 §2	REF-02	SRC-10
R8 library system	REF-02 §1	REF-02	SRC-10, SRC-19
R8 code tables (enums)	REF-02 §3	REF-02	SRC-10
SAP PM field mappings	REF-03 §3	REF-03	SRC-31
SAP upload templates	REF-03 §2	REF-03	SRC-32, SRC-33, SRC-34
SAP transactions	REF-03 §6	REF-03	SRC-30
Quality validation rules	REF-04 §8	REF-04	SRC-11, SRC-24
QA checklists	REF-04 §10	REF-04	SRC-23
OCP pain points	REF-05 §2	REF-05	SRC-01
MVP scope & timeline	REF-05 §3-6	REF-05	SRC-01, SRC-02
Data requirements (15 cats)	REF-05 §5	REF-05	SRC-02
Tech stack (full vision)	REF-06 §2	REF-06	SRC-50, SRC-52
UX/behavioral design	REF-06 §4	REF-06	SRC-55
10 AI use cases	REF-06 §3	REF-06	SRC-56
Work instruction templates	REF-07 §2-3	REF-07	SRC-40, SRC-41
Work package naming	REF-07 §4	REF-07	SRC-11

Topic	Quick Answer	Summary Doc	Full Detail
Material management	REF-07 §5	REF-07	SRC-10
Architecture & build plan	BP-01	BP-01	-
Project status	GOV-03	-	GOV-04
Competitive landscape	REF-06 §5	REF-06	SRC-57, SRC-58
GECAMIN competitor analysis	REF-10 §2-4	REF-10	SRC-70 to SRC-82
myRIAM SYSTEM comparison	REF-10 §2	REF-10	SRC-70
AI in mining maintenance	REF-10 §3	REF-10	SRC-71 to SRC-77
Asset Health Index	REF-10 §3.4	REF-10	SRC-78
Change management (SOMA)	REF-10 §3.6	REF-10	SRC-80
ISO 55002 compliance	REF-09 §2-3	REF-09	ISO 55002:2018 standard
ISO 55002 gaps	REF-09 §5	REF-09	-
Neuro-Architecture (behavioral UX)	REF-11 §2-3	REF-11	SRC-55
Cognitive load design	REF-11 §2 Pillar 4	REF-11	SRC-55
Psychological safety design	REF-11 §2 Pillar 5	REF-11	SRC-55
Behavioral nudges	REF-11 §2 Pillar 6	REF-11	SRC-55
Bias mitigation (UX)	REF-11 §3	REF-11	SRC-55
Strategic recommendations	REF-12 §2	REF-12	All sources
Competitive moat	REF-12 §2 Rec 10	REF-12	REF-10, REF-11
Phase 0 plan	REF-12 §2 Rec 9	REF-12	REF-05
Risk assessment	REF-12 §4	REF-12	REF-10
Asset Health Index	<code>gemini.md</code> §10.1	REF-12 Rec 4	<code>tools/engines/health_score_engine.py</code>

Topic	Quick Answer		Summary Doc	Full Detail
KPI calculations (MTBF/MTTR/OEE)	gemini.md	\$10.2	REF-12 Rec 8	tools/engines/kpi_engine.py
Weibull failure prediction	gemini.md	\$10.3	REF-12 Rec 6	tools/engines/weibull_engine.py
Multi-plant variance detection	gemini.md	\$10.4	REF-12 Rec 7	tools/engines/variance_detector.py
CAPA / PDCA tracking	gemini.md	\$10.5	REF-12 Rec 8	tools/engines/capa_engine.py
Management review (executive)	gemini.md	\$10.6	REF-12 Rec 8	tools/engines/management_review_engine.py
Expert cards (behavioral UX)	gemini.md	\$11.1	REF-11 Pillar 2	tools/models/schemas.py (ExpertCard)
Ipsative feedback	gemini.md	\$11.2	REF-11 Pillar 5	tools/models/schemas.py (IpsativeFeedback)
Completion progress (Zeigarnik)	gemini.md	\$11.3	REF-11 Pillar 6	tools/models/schemas.py (CompletionProgress)
Multi-agent architecture	gemini.md	\$7	-	agents/ directory
Agent tool wrappers (62 tools)	gemini.md	\$7.4	-	agents/tool_wrappers/server.py
4-milestone workflow	gemini.md	\$7.3	-	agents/orchestration/workflow.py
Milestone gates (approval flow)	gemini.md	\$7.3	-	agents/orchestration/milestones.py
Agent system prompts	gemini.md	\$7.5	-	agents/definitions/prompts/
FM 72-combo tool	gemini.md	\$7.5 rule 4	SRC-09	agents/tool_wrappers/fm_lookup_tools.py
Session state management	gemini.md	\$7	-	agents/orchestration/session_state.py
CLI entry point	gemini.md	\$7.6	-	agents/run.py

Topic	Quick Answer	Summary Doc	Full Detail
GFSN maintenance governance	REF-13 §1-2	REF-13	SRC-90
GFSN FMECA methodology	REF-13 §3	REF-13	SRC-90
GFSN criticality (Alto/Moderado/Bajo)	REF-16 §6	REF-13, REF-16	SRC-90, SRC-93
GFSN improvement techniques (OCR, Pareto, Jack-Knife)	REF-13 §6	REF-13	SRC-90
Planning & scheduling weekly cycle	REF-14 §5	REF-14	SRC-91
SAP WO status flow	REF-14 §4.4	REF-14	SRC-91
Work package contents	REF-14 §5.5	REF-14	SRC-91
Planning & scheduling KPIs (11)	REF-14 §8	REF-14	SRC-91
Priority matrix (Equipment × Consequence)	REF-14 §3	REF-14	SRC-91
RCA / Defect elimination	REF-15 §2-5	REF-15	SRC-92
5P's evidence framework	REF-15 §5.4	REF-15	SRC-92
3-level root cause (Physical→Human→Latent)	REF-15 §5.6	REF-15	SRC-92
5W+2H simple analysis	REF-15 §5.2	REF-15	SRC-92
RCA solution prioritization	REF-15 §6.3	REF-15	SRC-92
Defect elimination KPIs (5)	REF-15 §7.2	REF-15	SRC-92
GFSN 6-factor criticality	REF-16 §4	REF-16	SRC-93
GFSN 5×5 criticality matrix	REF-16 §6	REF-16	SRC-93
RCA API endpoints	gemini.md §7	-	api/routers/rca.py

Topic	Quick Answer	Summary Doc	Full Detail
RCA service (create, 5W+2H, advance)	<code>gemini.md</code> §7	-	<code>api/services/rca_service.py</code>
Planning KPI snapshots (history)	REF-14 §8	REF-14	<code>api/services/rca_service.py</code>
DE KPI snapshots (program maturity)	REF-15 §7.2	REF-15	<code>api/services/rca_service.py</code>
Defect Elimination UI	REF-15	-	<code>streamlit_app/pages/17_defect_elimination.py</code>

DOCUMENT STATISTICS

Tier	Count	Purpose	Typical Token Cost
Tier 1: Governance	5	Always read	~2K each
Tier 2: Summaries	17	Read for most tasks	~2-3K each
Tier 3: Sources	52+	Read only for raw detail	~5-20K each

Optimal workflow: GOV-02 (this index) → GOV-01 (gemini.md) → relevant REF-XX → SRC-XX only if needed.

Estimated token savings: Reading summaries instead of sources saves ~80% of tokens per topic.

CHANGELOG

Date	Change	Author
2026-02-22	Session 8b: Added 5 routing entries for RCA/DE full stack (API endpoints, service, KPI snapshots, DE UI). Updated document statistics to 17 summaries.	System
2026-02-21	Session 8a: Added REF-13 to REF-17 (GFSN methodology + gap analysis), SRC-90 to SRC-93 (4 methodology PDFs), 18 routing entries for maintenance governance, planning/scheduling, defect elimination, criticality analysis. Document count: 17 summaries, 52+ sources.	System
2026-02-20	Session 7: Added 8 routing entries for multi-agent architecture (agents/ layer): tool wrappers, 4-milestone workflow, milestone gates, system prompts, FM 72-combo tool, session state, CLI entry point.	System
2026-02-20	Session 6: Added 9 routing entries for GECAMIN engines (HealthScore, KPI, Weibull, Variance, CAPA, ManagementReview) and Neuro-Architecture models (ExpertCard, IpsativeFeedback, CompletionProgress). Updated REF-09 compliance to 80%.	System
2026-02-20	Added SRC-09: Failure Modes Lookup Table (AUTHORITATIVE) — 72 valid Mechanism+Cause combinations. Marked as MANDATORY for all FMEA work. Added routing entry.	System
2026-02-20	Added REF-10 (GECAMIN cross-reference), REF-11 (Neuro-Arquitectura review), REF-12 (Strategic recommendations), SRC-70 to SRC-82 (GECAMIN sources).	System

Date	Change	Author
	Updated routing table with 17 new topics. Document count: 12 summaries, 48+ sources.	
2026-02-20	Initial creation with 48 documents indexed	System Pilot